

EE551 Mini-Project: Parking Spot Detector and Classifier

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Abstract— This mini-project explores the use of classical image processing techniques to identify parking spaces and determine their occupancy from static images. The implemented pipeline operates on grayscale data and applies filtering, edge extraction, line detection, and morphological processing to recover the geometric layout of parking slots. Parking regions are formed through structured grouping of detected features, and occupancy is inferred based on feature density within each region. Experiments conducted on images with increasing visual complexity show consistent performance in controlled scenes and gradual degradation under challenging conditions. The study highlights how traditional computer vision methods can be effectively applied while also revealing their practical limitations. (*Abstract*)

Keywords— *Parking spot detection, image processing pipeline, grayscale analysis, edge extraction, line detection, morphological processing, occupancy classification*

I. INTRODUCTION

Limited parking availability and increasing vehicle density have made automated parking analysis an important problem in urban environments. Image-based parking spot detection provides a practical way to analyze parking layouts and occupancy without additional infrastructure. Two common approaches exist for this task: learning-based models and traditional image processing techniques. This project adopts the classical image processing approach due to its lower computational cost, transparency, and suitability for simple systems.

The objectives of this mini-project are to detect parking space boundaries from grayscale images, classify each space as occupied or empty, and evaluate system performance under varying image difficulty levels. The implemented method relies on edge detection, Hough line analysis, morphological processing, and threshold-based operations to extract parking structures and infer occupancy. Experimental analysis highlights both the effectiveness of this approach in controlled scenes and its limitations when visual complexity increases

II. METHODOLOGY

A. Image Acquisition and Preparation

The system operates on a set of ten parking lot images categorized into easy, medium, and hard cases based on visual complexity. All images are processed in grayscale. To improve visibility under uneven lighting conditions, contrast enhancement is applied using adaptive histogram equalization. A Gaussian filter is then used to suppress noise while preserving edge-related features needed for later stages.

B. Gradient Computation and Edge Detection

Horizontal and vertical intensity changes are extracted using Sobel operators. These gradients emphasize structural boundaries within the image. Canny edge detection is applied to convert gradient information into clear and continuous edges, enabling reliable boundary extraction.

C. Line Detection Using the Hough Transform

The Hough Transform is applied separately to detect vertical and horizontal line segments corresponding to parking space boundaries. Morphological opening is used to eliminate small spurious detections, improving robustness against noise and clutter.

D. Grid Formation and Parking Spot Localization

Detected line segments are merged to form a grid structure representing individual parking spots. Minor alignment errors caused by perspective effects are handled through adaptive thresholding and intensity-based filtering, allowing consistent grid formation across different image conditions.

E. Occupancy Classification

Each detected parking region is analyzed using morphological erosion and dilation to isolate potential vehicle presence. Occupancy is determined using intensity-based criteria, where darker regions indicate occupied spaces and brighter regions indicate empty spaces. Results are visualized using color overlays for clear interpretation.

F. Pipeline Summary

Input Image → Grayscale Conversion → Gaussian Blurring → Sobel Gradient Computation
→ Canny Edge Detection → Vertical and Horizontal Hough Line Detection
→ Morphological Opening → Line Merging and Grid Formation
→ Erosion and Dilation → Parking Spot Classification
→ Visualization using Color Annotations
The pipeline relies on edge strength, geometric structure, and intensity variations to handle different image difficulty levels without extensive parameter tuning.

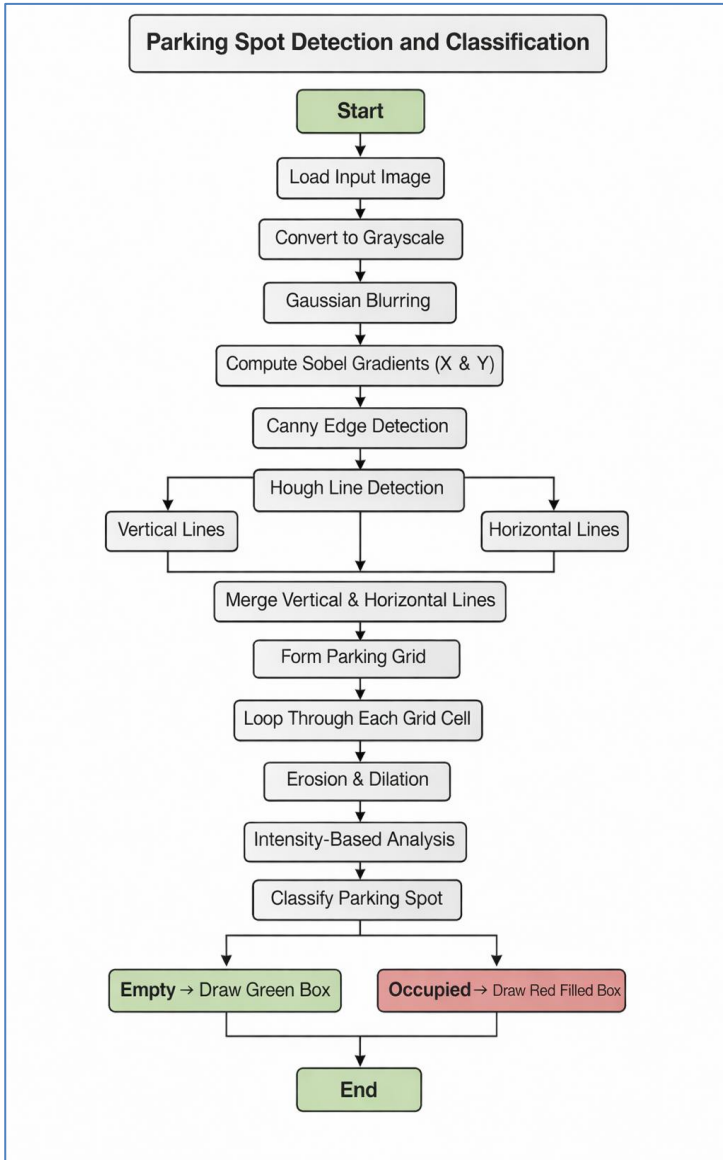


Fig 1. Flowchart of Pipeline

III. RESULTS

The performance of the parking spot detection system was evaluated using ten test images categorized into easy, medium, and hard difficulty levels. These categories were defined based on visual complexity, including factors such as lighting variation, shadows, occlusions, and perspective distortion. The results highlight how the effectiveness of traditional image processing techniques changes as scene complexity increases.

Easy Images: The four easy images demonstrated strong performance across all stages of the pipeline. Parking space boundaries were clearly visible, allowing the Sobel gradients and Canny edge detector to extract clean edge maps. The Hough line detection stage consistently identified both vertical and horizontal boundaries with minimal false detections. As a result, grid formation was accurate, and individual parking spots were well-aligned. Occupancy classification based on intensity analysis performed reliably, producing a nearly perfect overlay of detected parking spaces. Very little parameter adjustment was required.

Medium Images: The three medium-difficulty images produced satisfactory but less consistent results. Shadows, partial obstructions, and faded markings introduced noise into the edge detection stage. In several cases, the Hough transform detected extra line segments that did not correspond to actual parking boundaries. These false detections occasionally caused slight grid misalignments. Despite these challenges, most parking spaces were still correctly identified, and occupancy classification remained reasonably accurate, though some spots near shadowed regions were misclassified.

Hard Images: The three hard images posed significant challenges to the system. Strong perspective distortion, uneven illumination, and visual clutter reduced the reliability of edge and line detection. Vertical and horizontal line separation became less stable, requiring manual adjustment of merging thresholds to form a usable grid. In darker regions, intensity-based occupancy classification produced false positives and missed detections. Although acceptable results were achieved after tuning parameters, performance was noticeably less robust compared to easier cases.

Overall, the implemented image processing pipeline performed strongly on simple scenes, delivered reasonable accuracy for medium cases, and achieved usable results for complex images after careful adjustment of detection thresholds and other key parameters settings.

IV. DISCUSSION AND CRITIQUE

A. Analysis of Individual Processing Stages

Preprocessing: This helped to reduce the noise and stabilize edge detection, which was especially important for the medium and hard images where lighting variations and background texture were present.

Sobel & Canny Edges: These stages were effective in highlighting parking boundaries in most cases. However, in hard images with poor contrast or strong perspective effects, the extracted edges were often incomplete or broken, which reduced boundary continuity.

Hough Lines & Morphology: These were essential for building the grid. The vertical pass performed better than the horizontal one at detecting aligned parking spots. The opening operation did a good job removing small artifacts, but in complex scenes we sometimes merged lines that should have stayed separate.

Erosion, Dilation & Occupancy Detection: This gave us effective classification in easy images, but intensity variations in medium and hard images caused mistakes here and there.

B. Performance Across Image Difficulty Levels

For simple images, the pipeline operated reliably with minimal parameter adjustment. In contrast, medium and hard images required tuning of line detection thresholds and grid merging rules. Perspective distortion in angled views needed additional care to maintain correct parking slot alignment.

C. Observed Failure Cases

Medium-difficulty images: An increased number of incorrect detections occurred due to faded road markings and shadowed regions, which interfered with reliable line extraction.

High-difficulty images: Strong perspective effects and surrounding clutter, including nearby vehicles, poles, and irregular pavement surfaces, further increased detection errors and reduced grid stability.

D. Feasibility for Embedded Deployment

The proposed image processing pipeline is lightweight and well-suited for execution on embedded platforms with limited computational resources. Most of the processing cost comes from edge detection and Hough line analysis, which remain feasible for low-resolution inputs. The step-by-step nature of the pipeline also supports efficient sequential processing, making it practical for resource-constrained systems.

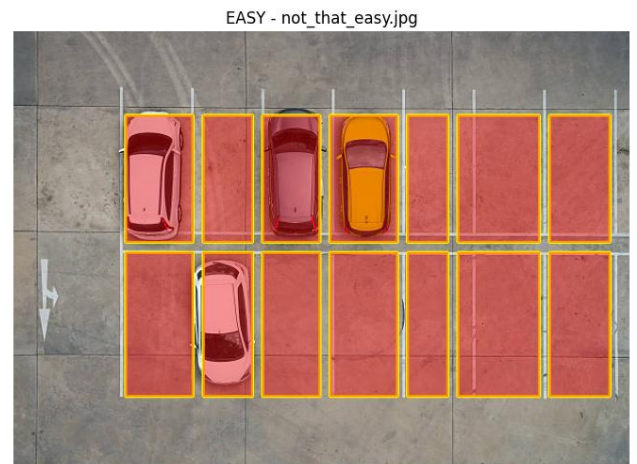
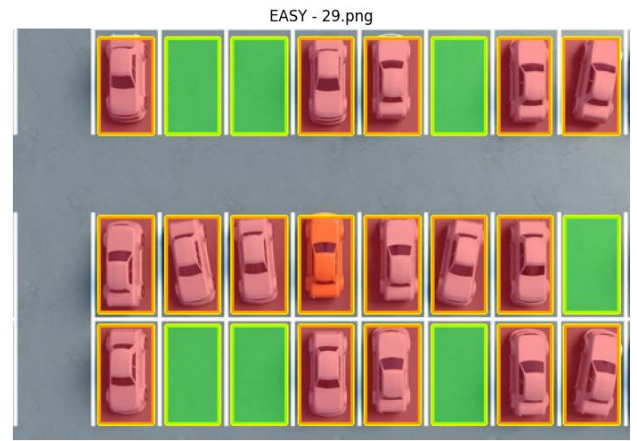
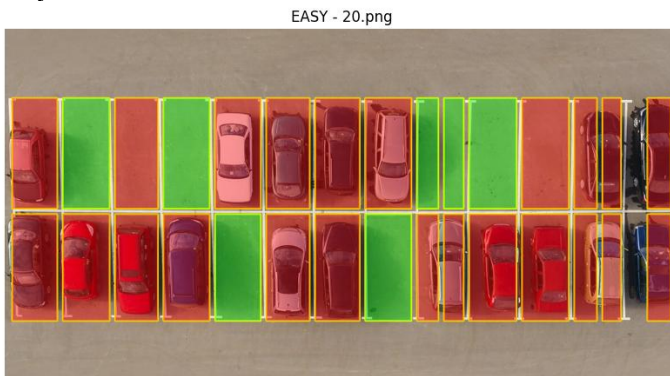
E. Overall System Evaluation

The results demonstrate that classical image processing techniques are effective for detecting parking spaces under stable visual conditions, but reliability decreases as scene complexity increases. The approach is particularly suitable for low-resource environments and provides clear interpretability compared to learning-based methods. Future improvements may include handling perspective effects more explicitly and refining occupancy decision logic to improve robustness in visually challenging images.

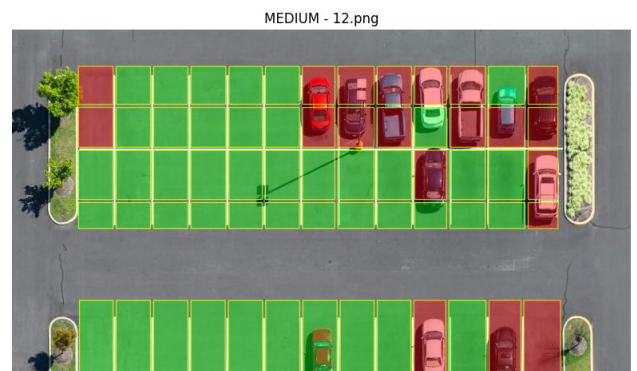
V. CONCLUSION

This project developed and evaluated a classical image processing pipeline for parking spot detection using a set of ten images with different levels of visual difficulty. The system produced accurate parking boundary detection and occupancy classification for simple scenes, while performance decreased for medium and hard images due to factors such as shadows, perspective distortion, and surface markings. The analysis highlights the role of each processing stage and demonstrates the computational efficiency of the approach, making it suitable for embedded applications. Classical techniques continue to be a practical choice for real-time parking analysis when low cost, transparency, and simplicity are required. Future enhancements may focus on reducing false detections and improving grid alignment to achieve better robustness in complex environments.

Easy:



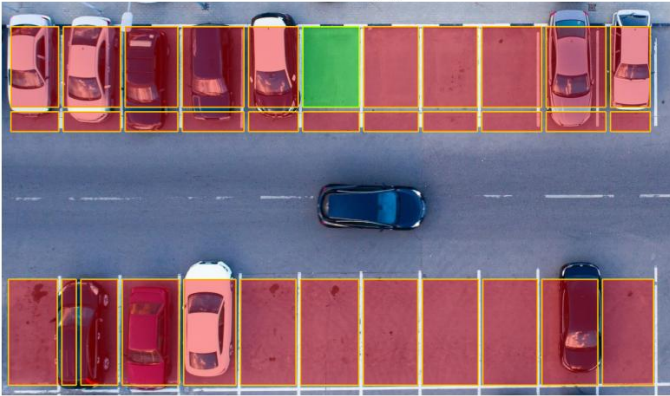
Medium:



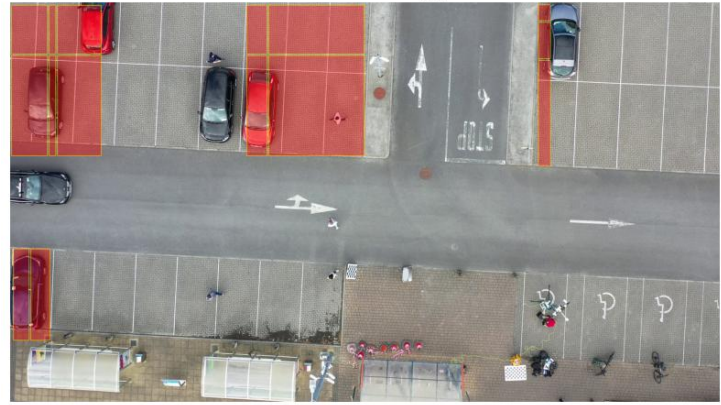
MEDIUM - 30.png



MEDIUM - 31.png



HARD - frame600.jpg



Hard:

HARD - 21.png



HARD - frame1.jpg

